

ActInf Livestream #021.03 ~ John Boik

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Duration: 2:08:53 | **Uploaded:** Unknown **Transcript method:** auto_caption

okay hello and welcome to active inference lab this is the third in a ongoing series of discussions between i guess active inference lab and john boyk i'm daniel i'm a postdoc in california and i'm going to be kind of working through the jam board as we've done in these other sessions while john goes through his paper some of the implications we can also raise up some questions from the live chat so people are totally welcome to um leave questions in there wait actually it's already sort of messed up or not sure i hope that it will work it looks like it's okay we're just sort of dealing with the streaming tech and i'll play with it once you start talking jon but um yep i hope this uh is a fun discussion because the other ones have been awesome as well so i'll flip over to slide 14 from our second slideshow and let you pick up with some points from last time and uh let you know if anything is awry okay sounds good thank you my name is john boyk i'm a courtesy faculty at oregon state university environmental science graduate program and this series is about our series of discussions is about a series of three papers that i recently published in the journal sustainability and the topic is societal transformation um i think somewhere in the middle of this talk we'll show some links to those papers and and to my website where all this information can be found uh so we're do you have the slides up you want to put up slide 14 we're just going to jump in we're going to jump into halfway through the last talk we had a bunch of slides that we didn't get to so we're going to try to get to them um now yep i'm there so i'll be following along but you just won't see it but we're good to go oh i won't see it okay all right okay all right so uh slide 14 is uh is a list of 14 desired data that i had proposed in this second the second paper part two uh i should probably say what the title of that series is the title is science driven societal transformation parts one two and three so we were talking about part two last time which was on um motivation and strategy and we're going to continue with that now um so in that part two paper i had listed 14 desired data for what any what would be what maybe could or should be in any r d program aimed at societal transformation um throughout the series i touch on all of these fourteen um i would just like to make clear that what i'm proposing the series the three-part series is itself a proposal for an r d program it's kind of an informal

proposal to the world about a program that could be started and it is possible that other people have other groups have other ideas of what kind of r d program you know could be constructed and you know it could be that in the future there's numerous uh r d programs that are in operation that are all looking at this topic of societal transformation and i wanted to offer a list of uh of points or topics that that each of them might address and that way at the very least we could compare programs by comparing these for 14 points um i just we we covered we went through him last time so i'm not going to go through him again but i just wanted to say a couple of things uh daniel i remember from our last conversation you had said something um you know getting a a bunch of scientists together and maybe they won't agree on you know different aspects of this and there you know could be some arguments for one point over another point or one approach over another approach and that of course that is true and that's fine and that's good there should be a variety of voices uh heard and considered in in any r d program but i i did want to make the point that the r d program that i'm proposing the the concept is evidence-based science-driven r d program so um suppose that uh well let me say that that an evident evidence-based program is different than a popularity-based so it's not just that you know everyone gets to vote for what they want and it'll and and the r d program would go off in in whatever direction everyone votes for i mean in in a sense that's true because how else would you govern things but in another sense it's not true because there has to be some kind of science process i mean we have to be addressing with the with a design program we have to be addressing the the science-based aspects of it so um and that is largely driven by fitness purpose and fitness so the so all right um let's see yep just continue going ahead thank you okay we're good yep okay all right uh i was saying that um that there could be multiple p multiple uh uh r d programs proposed by different groups driven by different groups but i would i would expect that they would all be based on some kind of science-based evidence approach so while while each r d program might have a slightly different way of measuring fitness those discussions would be based on scientific evidence and reasonable you know reasonable evidence as opposed to just a popularity we're good yep working with tech thank you everyone for standing by when it kind of flips out for a second we'll try to restart it like as soon as possible but that's why we use the asynchronous modes of work because we're really lucky when we get to have synchronous time with the authors or with anyone but the and the updates on the comments are what really matter for this kind of a project not someone's internal understanding or just a dialogue so thanks for bearing with us everyone go ahead john okay just to make that last point is the r d program that i've proposed is really it's

really a meta meta framework for learning new systems new system designs so the idea is that the the r d program would look at a wide variety of of designs people people it would provide a way a means for evaluation of different designs that different groups might come up with so that's all i wanted to cover on that topic and then on to new information so yep okay we'll be able to do it right okay here is this this is going to work give me one second all right thanks everyone for holding by i hope that you enjoyed this section the silence to think about these ideas okay so now it's called making lemonade out of yeah so now i will spend some of the time when you're talking john to just continue like trimming and stuff but we actually um are back in the game so please continue on 5 15 and 2 and go from there and then continue on so sorry about that first 15 minutes of fun now go for it just one little thing my screen my picture on my computer is quite dark but i don't know if i'm coming through on everyone else's computer looks good and i'm going to fix it on the other computer okay all right all right so slide 15. uh uh i was saying before before the interruption that uh this r d just a few things on the rd program it's a it has a as i envisioned at a 50-year horizon so this is um this is a careful movement towards societal transformation it's going to take a a a long time but um there will studies will be if it were funded if it moved forward studies would start right away field trials would follow in in the near term and those communities that choose to participate in field trials would see you know benefits far faster than than than in other communities that are waiting so this is a 50-year horizon towards a global scale or no near global scale transformation so it's going to take a while to to achieve change at uh you know at that huge scale um uh i also want to mention that the program involves the public in a variety of ways this is i envision this as a partnership between the public especially between local communities and the science the global science community so there would be field trials there would be a variety of studies and the public can get involved in a variety of ways there would be lab lab studies there would be surveys there would be the ability for the public to in input their ideas and their their wishes and desires into the program so hopefully this would be a a conversation a multi-way conversation between the stakeholders that are involved and uh on the science side uh i mostly want to make the point that the program could involve nearly every academic field that's out there literally in a to z and at the bottom of this slide i just list a few of the many fields that would be appropriate here [Applause] certainly some of the the science fields have been talking about complex science other fields like that but also education media literature linguistics like you name it and there would be a place a need for participation in this project okay next slide 16. okay we're good yep all right uh so um you know the idea that the

science world would be actively participating in a program for societal transformation this is you know in a sense this is rather bold and new uh novel um there there obviously are programs aimed at the study of transitions and and and developing the knowledge base for societal trans uh some kind of a societal transformation because many in the science world and the public policy world understand that bold change is necessary if we're going to uh you know uh avert the avert problems that we're facing due to climate change uh bio biodiversity loss and a number of other very serious social and environmental problems so big change deep change is on many many uh people's minds but uh science uh the role that science has that many people envision for science is more of a knowledge is more than the traditional science world the knowledge gathering knowledge product production that would support the public and and politicians and choosing new policies and various things so the idea that science would be involved in the actual r d of making new systems designing new systems testing new systems that's more of a what is being called second order science and that is fairly new in the science world so this is uh this is breaking new ground for uh force for the science world so i i have a bullet list here of various reasons why the science community might want to be involved in this so i'll just go through a few of these real quick um first societal the idea that society is a can be understood as a organism a super organism if you will and a cognitive organism at that is is timely and important and offers numerous opportunities to advance basic science knowledge so um so while in in participating in a program of societal transformation a great amount science knowledge would be generated uh next is systems are societal systems are sophisticated so uh the the skills that are offered by the science community are extremely important uh they would have to be computers computer simulations for example and a variety of other very technical uh problems that would have to be solved in order for this for new systems to really work and for the design process to work so uh for that you know the science community is uniquely uniquely qualified to contribute to those effects uh the goals of transformation are ambitious this is the third bullet and yet the time available for action is short so this has to be not tinkering but moving in a direct and focused way towards societal transformation and the developing new systems and the science community can help with that to keep it all on track and moving forward i've already mentioned the second order science threshold threshold to a new way of doing science and that can be useful and involve science more in the public sphere in the future and the next is that the science community is trusted by the global public so uh and is a bishop and then is a position to help focus public attention on pressing problems and and solutions next to the last one is the science community can offer a

global perspective and global coordination so what we would not want are a whole bunch of experiments to occur for the r d program and then to be forgotten and never picked up again or never integrated into the body of knowledge after that so the science community can offer a central in a sense a central focal point a repository for data and for assessing data and for assessing the many experiments that might occur in such a r d program and lastly the science community can act as a powerful ally to the public uh in promoting and ensuring that transformation moves forward so i hope that i hope that the science community will find interest in this kind of r d project that's really the purpose of the whole series is to offer the the idea to the science community that we could do this together this could be a very important and timely uh service that the science community offers to the world all right on to 17 all right so and i said this before but if i were the smartest person on the planet and i came up with a fantastic new design for societal systems and and for viewers that are new by societal systems i mean economic systems financial systems governance systems educational systems legal systems and the like um if i were the smartest person i came perfectly designed societal systems if there was no way to implement them practically then my effort would have been waste wasted so any r d program aimed at societal transformation has to have some kind of viable strategy for implementing any new designs that might be created and uh in this in the proposal in the series i i i offer a theory of change and a strategy for change that i call local global viral and the idea there is the local global is the idea that local that much of the work would occur at in local communities and by local communities i mean volunteer communities volunteer individuals who would develop who would uh participate in the in the in the testing of new systems and and even the design of new systems uh in cooperation with and partnership with the global science community so we have the local global connection there and and the viral connection is such that the concept of the of the i i use what i call a club model and the concept of the clubs is that clubs would be implemented locally but on a volunteer basis as i mentioned and they would be designed such to produce substantial benefits for the participants and if that is if they're well designed and indeed the participants obtained substantial benefits then the concept of the the technology and the concept for the clubs and the club model and the and the systems that they embody would quite naturally spread from one community to the next uh if a if a club starts in a in say in a city just a not not as a city but just even a small fraction of a city that that could be a club any like a club could be any uh maybe perhaps a thousand people or more so just a subset of a city could start a club and be part of this r d pr or the whole r d program a field trial could happen uh uh via a

club and if the club really does produce benefits then the next city over the next community over is going to have an interest in starting it also because the benefits are are substantial and at the same time a club can grow in the participation participation rate in a local area could grow so clubs can grow in size and they can replicate into new locations and that sets the stage then for viral growth and that it would be how the the societal transformation and the use of new systems could spread to near global levels over a period of the 50-year horizon so uh clubs are designed to basically uh out compete native systems by native systems i mean just the you know the the the legal economic governance systems that we are using today these new systems will obviously be designed for a purpose and that purpose is to benefit and improve the quality of life and the security of of participants and uh if if if they achieve that they will out compete native systems for the public's attention engagement trust and respect so that is the that is what they should do and uh and if they do then then the their use would would quite naturally spread in a viral fashion now the clubs i just want to say about the love the club level so we're not talking about say a family making changes that's like this isn't really family focused the whole project it's not really focused at the business level like a particular business a new a new business model for example it's not focused there and it's not focused at the city level so a new city charter or a new new way the city can operate to be uh to improve somehow it's situated in between the city level and and the kind of family slash business level right and i think of that as the goldilocks level uh if a club was smaller this i would say this this level this in between level the middle ground sometimes i call it is is the smallest level of organization that would allow a club to be viable testable self self-replicating and effective so anything smaller than a the club level uh you could imagine say if we're just talking about a business well then that business has to compete in an environment against other businesses to survive and if those other businesses are you know not on the same in the same you know doing the same thing on the same program they might have advantages that the new business model might not have at the same time if we went to a higher model say a city level uh trying to change things at the city level well that's suddenly uh really expensive and difficult especially for a large city so there would be a lot of political blowback for trying to change things dramatically at the city level and the club level is such that that you could implement you could you could implement new systems without having to deal with uh the need to to win elections or uh uh you know get approval from any public body right so this is so a a thousand like-minded individuals can just start a club and no need to no need for anyone else in the community to join in they can do it all on their own just as is so that's the idea is to reduce the

barriers to to to participating in a club in in new systems and uh and have it be such that the that that new systems could virally spread over time if they if they show their benefits okay so just to make the point then when i'm talking about societal transformation the new designs that we're coming up with i'm talking about the designs for a club not the designs for a city not the designs for a county not the designs for a state or for a nation but i'm talking about societal systems designs for a club and the and the club will then demonstrate to the world in field trials whether this whether these new systems work well as as advertised or not nice good point because sometimes when we're talking about these complex multi-scale systems it's good to make it clear that we're talking about policies affordances capacities that are at a certain level of analysis we're not invoking a nation state or something like that no we're we're doing something we're working at what uh a workable level you know this is focused at a workable level this can happen there's really no reason it couldn't happen and uh obviously getting the initial funding for the project is a big deal but once the initial funding happens i don't see why this could not move exactly as or you know close to close to what is planned here um yeah next slide uh is this the endgame vision 18 is that where we're at yep yeah okay so um you know where is this all leading i've already said there's a 50-year horizon uh aimed at at societal transformation on a near global scale but but the real goal obviously the real goal in this is is for society's humanity to to use systems that are beneficial to them that are most you know most beneficial or widely beneficial to them i believe that i should say there's there's good reason to expect that a well-designed r d program that engages local communities and the science world could come up with designs that are substantially better than our current ones i mean if you if if the goal is to uh is to increase quality of life and reduce uncertainty if that was how you were judging the design of societal systems then i think you would have to say that the current systems are are failing they're they're terrible in the sense that we now face uh you know enormous stress and strain due to climate change due to biodiversity loss due to due to a host of interrelated problems that all all of which were preventable all of which have been talked about in the science world and in other communities for the last several decades and all of which had we taken action we would not be under the risk that we are right now so and so if nato if native systems have given us this extreme a world of extreme risk then i think it would be fair to say that native systems are failing humanity and it would be hard to imagine that a that a sophisticated science-based effort with some of the best minds in the world you know focused on this this topic could not do far far better than current systems so just to connect it to one note from game theory it's that the strategy has to be resilient along the way

so it needs to have that kind of like introductory dynamics and getting off the ground and bootstrapping or not relying on lateral connections amongst clubs it has to be sort of an end in and of itself and then it also has to be a resilient mixed strategy at 50 percent it has to be a resilient dominant strategy and so flexibility to play different strategies at different um phases of the uptake is critical that's why you've laid out a timeline as well yeah yeah um and that's also you know kind of related to that to your point is that there could literally be hundreds or thousands of experiments going on around the world in different communities of of different system designs and different asp different aspects of a particular design so the so you would want to be very flexible very adaptable we're learning as we go as a matter of fact uh and would up you know would change the change of the strategy slightly as we go and change the focus slightly as we go adjusting the circumstances and and starting small and growing organically helps you to do that as opposed to starting big and perhaps failing big so in the end game vision the idea is that in the future humanity is using systems that really serve humanity uh uh as we've talked about many times now it's just um the view here is societal systems as a cognitive architecture as a way of facilitating societal cognition and obviously we would all like our societies to be good at thinking good at learning good at adapting good at making decisions so that we can all enjoy the benefits of such so the endgame vision is perhaps i mean there's here's a couple of scenarios for the endgame vision the r d program moves forward we demonstrate useful designs where that give substantial benefits uh to communities and the existing native systems uh slowly uptake those ideas and blend those ideas into their into their systems and then in the end you know where you the native systems become much better so that would be great that would be fantastic and another scenario is that native systems fail to adjust and change and up uptake these new ideas they uh and if that were the case since the the new systems are designed to earn the public's trust and respect and attention um they would it would seem reasonable that they would out compete native systems for the for the public's trust and respect and if that's the case then you know people would be using these new systems and relying on them more so than native systems so those are just two scenarios but you could you could imagine any num you know a number of other scenarios that might occur from such an r d program but uh the either way at least with these two scenarios either way the end goal is reached of of of you know a large majority of the human population using systems that actually serve them and serve them well slide 19 yep okay so a little bit more in clubs so by the by the time the first field trial occurs in the first system and this could you know potentially uh the field trial the first field trial could occur maybe anywhere in the world

practically but by the time the community participates in a first field trial it should be really clear what the expected outcome is right a community does drive its systems a community makes decisions day to day the of of what you know what kind of uh you know about services about waste control about energy generation about all kinds of things a community makes its own decisions but the general trajectory towards uh improved quality of life improved well-being improved security improved economic security uh reduced income inequality like all of those things there should be very few surprises as to what occurs in a field trial it should be there would have already been numerous simulations that would have occurred before uh before our first trial first field trial already uh before it even begins there would have been all kinds of user testing that would have occurred there would be all kinds of game playing with new systems that would have occurred so by the time the first field trial starts this should be a you know this should be a trajectory of exactly what is uh what is predicted and so there should be few surprises for the field trials now obviously that doesn't mean that something you know there might not be hiccups or whatever but by and large there should be very few surprises the community should know really what what they're getting into how the system operates and what the expected benefits are in each year of participation second item is that that while the initial field trials might be really focused at the individual club level as the r d program progresses there would be an effort to connect clubs via networks so that clubs themselves are participating in some kind of cognitive uh operation um cl as a community of clubs that also needs to make decisions to learn to adapt uh to make decisions together so the the idea is that some of the same kind of approaches and technologies that are implemented at the club level could also be implemented at the network level so that the entire uh the entire you know range of clubs and the individuals within clubs could all be um participating in some kind of optimal cognitive system right making good decisions and look and and opportunities for learning uh next slide um so why why why why the why the club model uh i've already mentioned a few things uh it's in the goldilocks level that allows for replication and and and viral spread but there's a few other uh advantages that i would like to mention um first we're talking about when you think about bold change revolut violent revolution is one kind of bold change so we obviously are not talking about revolution we're not talking about any kind of violence whatsoever we're talking about improving the well-being of communities at every step along the process so this is non-violent and that's one part another part is that success is measured and success and progress are measured throughout the throughout the process so not only might you be doing better but that would that would be validated through a

variety of metrics you know there would be this is an experiment you know this is the science world is engaged in this ongoing experiment to develop new systems so data would be gathered and you and everyone would know that you're doing better if you're doing better right that we would know if they're working or not maybe that's a better way to say it fitness the evaluation of system fitness and the quality of life for community are central to this whole project again because the clubs are operating in that middle ground there's that minimizes political opposition opposition there's as i said any group can join a it can form a club so that you don't need a city's approval for this you don't need a nation's approval for this is this is just a a civic club so a civic club can start nearly anywhere in most countries or most democratic countries and so you don't need a you don't need a vote and the re this is really important because i am i am speaking to the global science here and i'm inviting the the global science community to participate but i don't expect that everyone all scientists will will be interested in this nor do i expect that all individuals in the general population would be interested in in participating in a field trial or or anything like that as a matter of fact i would expect that most most people you know most people in any given city will have no interest in this at all but that doesn't matter because again all it takes is a thousand people in in your city or your area to start a that club shows benefits it quickly becomes clear to others that this is a this is a very useful uh this is a very useful program incomes are rising incomes are stabilizing quality of life is improving the club feels more empowered to make decisions that affect its that affect its future the there's a stronger sense of um there's the ability to to a better ability to deal with problems as they arise and find long lasting solutions so on on numerous levels quality of life and environmental quality should increase information should increase transparency transparency should increase democratic participation should increase and those are all really you know attractive attractive characteristics if if indeed a the club model provides that so uh if that's the benefits then then again the there's the idea will spread from one community to the next and grow within different communities and it's not necessary that that that a large percent of the population jumps on board on day one it's just it's just not necessary there must be i i would argue that even today if we looked around looked around there would be a community of a thousand people who would be interested in participating in a field trial and the first field trial of course is still some years away the program isn't even funded yet so by the time we get to the point where we really need a community to run a field trial i i imagine there that will be quite easy as a matter of fact i would expect numerous communities to offer themselves as participants for the first field trials um this just a few more points here but the city whoever in

in whatever city the first field trials occur they're likely to have beneficial spillover effects so uh you know tax revenues might rise for example uh other other spillover effects from the club might help the city at large so you know maybe even cities would welcome such a thing if it's going to improve the quality of civic life uh for those who partici for those who participate and for those who don't participate um and this this club model is opposed say going big to changing the for example the u.s economic system or the global economic system on you know in a on day one or in a short amount of time this uh this local global viral strategy is far less cost costly far less risky and far more flexible we're talking about starting small bootstrapping or growing and that all can be done on a on a practical budget as we'll talk shortly and a lot and as i mentioned it allows for parallel testing uh so numerous field trials in numerous countries and or cities could be happening all the same time and as far as cost goes what you know what would be the cost well it's a little hard to say because the program isn't really developed yet so but this is just a kind of a guess on my part but i'm i would think that in the first few years the cost might be less than 10 million dollars a year to get this started and then in in sub subsequent years still in the first decade the cost might be on the order of tens of millions of dollars now that is you know in a certain in a certain sense that's a lot of money but in in a certain world that's a lot of money but in another world that's peanuts so this is not this is about the same cost as a modest size or a good size startup uh you know so that this level of funding is available in the world uh it's this is not a crazy cost and uh i i argue that you know with this modest cost we could really change the direct trajectory of humanity here okay that's it for part two oh that's it for part two okay all right so uh so we have we will continue this uh talk about part two in our next jam board so i'll add the new link and we're now on slide two of jamboard three okay all right good um uh so the slide two is uh is just where you can go for more information so if you'd like to know more about the proposed program if you would like links to all the articles and links to all the previous videos that we've made and that i've been involved with you can find them all on [Music] principledsocietiesproject.org all right slide three um so there's a few remaining topics from uh part two that i'll talk about now um some opportunities outside of clubs a little bit about fitness metrics and then some final thoughts on part two yep so we've already talked about opportunities in clubs which could include for example increased income um and uh higher quality of life and a bunch of other material benefits material and social benefits but there's also some opportunities that could occur for those outside clubs and as i mentioned clubs sit at the midway level between the kind of family slash and the city level and both of those levels are already well developed and you

know there's an enormous amount of revenue attached to both of those levels the middle ground is not yet organized but as it organizes i would imagine that revenue opportunities are you know similar to those at the city level and those at the at the family slash business level so uh the i the the the the r d program would help to do to organize this middle ground the club and some of those there's a little bit of evidence that that such organization is already happening for for example uh is it called what is the app called the neighborhood app it's the name escapes me right now but there's a there's an app that is widely used and is used by millions of people in the us at least about communicating within your neighborhood and sharing information within your neighborhood right so that's already a big business that that app alone is a big business cities are also developing community energy supplies community solar and that's just another way that this middle ground is is organizing and if the r d programs goes forward there would be a tremendous amount of organization happening within this middle ground so that opens up opportunities for consultants for example most of the work and developing the r d pro program the the tools and technologies needed for field trials for new systems that would mostly be open source you know publicly available uh knowledge information but it could be that that businesses consulting businesses or other businesses could provide services to new clubs could provide these could be training services they could be you know like internet services or any other kinds of services that would help clubs get started and uh i imagine that could be a big business as we're as we're as this moves forward maybe a good model for that is red hat incorporated which uh which uh provides services related to the linux operating system provide services to businesses related to linux linux itself is an open source product or it's open source technology and red hat which is a enormous company i don't know if it's a i see a billion dollars in tax revenues recently so this is a very very large company operating on top of an open source stack in a sense and the same could be uh the same operation similar opportunities could occur for for for businesses who want to work in this consulting or service providing space another opportunity is for cities and other governments because of tax revenues later in i don't know if we'll get to it in the next session but maybe in the in the next next session section of these discussions we'll get to the uh the simulation model that i published on what i call the the lita framework local economic direct democracy association framework uh that is that is that simulation is just illustrative of a potential uh kind of a prototype new um societal design uh focused on the economy focused on an economic system but in the simulation that occurred the simulation was for eugene oregon using data census data for eugene oregon and uh the simulation occurred over a 28-

year period but by the end of that 28-year period um let's see i headed down there just a few words about what that provided but billion in tax rate right right which was more than double the initial tax revenue so potentially this is a you know could be in a sense kind of a windfall for uh local governments city governments and community governments uh for for tax revenues and then finally there's an opportunity for large investors and that uh you know we are looking at a future world that is going to be very stressful because of the the implications of climate change the effects of climate change uh biodiversity loss and social other social environmental problems so if it happens that uh that there's a shortage of good investments to protect your money the club model and the club process and the r d program might become they might be seen as safe havens for parking money until things uh you know improve and if that's the case i could imagine that some institutional investors who who desire to make social investments or who see the wisdom of making social investments could invest in this whole process and of course that would provide funds for developing the developing the club model helping the club model to spread and um and grow um yeah who knows who knows what is going to happen in the stock market as clim the ravages of climate change approaches and maybe the maybe this maybe this club model idea could really actually help large investors by providing a safe haven uh for their investments and i'm talking social investments i'm not talking um highly profitable investments that some might be used to but i'm talking about uh social investments where it's invested in a low expectation of a profit okay slide five that's cool uh okay so just a little bit fitness metrics we've been mentioning them all through these talks i just wanted to say that they serve several purposes for example they are they are this fitness metrics the information gathered about fitness metrics so for example just to name a few fitness metrics you know could be uh family income could be education levels it could be water quality air quality i mean you name it that would be you know potentially uh hundreds or thousands of of of topics to collect information on so this in this information could help a club assess its own status how am i do how are we doing you know we need information to answer that question how are we doing well we can say that we are doing blank you know we're doing so well or so poorly or whatever the case may be on education on environmental quality on economic security on all these different issues um second is this information helps clubs to solve we have for example the how do we how do we maybe the problem is focused on economic security how do how do we solve this economic security problem well we need information we need to know uh how we're doing right now how are we doing last year so that we can know we can make good decisions for solving these problems that we're facing right

now metrics are also useful for assessing club and system design again clubs are built for purpose and are supposed to function successfully for purpose and achieve the aims that they were designed to achieve and all of that to assess club fitness and to to assess how a club how a club design is doing requires information another use of information is just informing members so everyone has questions about you know just day-to-day questions in their life you know what career should i choose or or you know what where is it where can i buy fresh strawberries today or whatever you know whatever information that individuals need in living their day-to-day lives and running their businesses day-to-day the information gathered by a club could help them to make better decisions just in their own personal life and then lastly the data assessed would just be a windfall for basics basic and applied science this were you know in a sense we're talking about a repository a a large repository of longitudinal information that follows individuals and families and businesses through their life and uh and how their how they're doing their quality of life as as decisions are made and as the as the as the club makes decisions as they make individual decisions um that information would just be gold for understanding uh you know societies better understanding clubs better understanding health better you know than that information there's only a few small repositories like that today so we're talking about just a really a gold mine for for science and a basic science and applied science if this moves forward obviously there would be you know personal you'd have to protect personal data in all of this but that could be done and just a few words about how this these fitness metrics and data this data gathering system would be different from what's already done well you know already for example the census collects information and the sustainable development goals sdgs uh are proposing a you know a collection of data at the national level um that would be and all of the all of that data for the census and for sdgs it touches on some of the same topics like quality of life and education and economics and so some of the large topics are similar but the but there would be many many differences between this kind of data collection and uh and the existing programs so just a few differences this would be a larger and more detailed set of it'd be collected and managed at the community level not at the national it would be about local conditions so specific for local conditions rather than say national conditions forecasting other kinds of models are integrated into this repository so this would be a you know an intelligent uh the aim is here an intelligent data gathering system it would be part of uh you know forecasting metrics into the future and things like that uh and the obviously the club is designed to serve a different set of purposes and in particular purposes related to community cognition societal cognition lastly clubs

would collect information on a far more frequent basis so uh you know some some survey questions are asked every few years sometimes sometimes every year sometimes every few years sometimes every five years and the sdgs i think are collected mostly on an annual basis so uh this is on a perhaps you know some some data would be gathered minute to minute or hour to hour or day to day so there would be a far richer and higher quality data set that would that communities would generate yeah and the basic idea is the community needs to know what's going to happen to me like a community needs to know what's going on where did we come from what's going on where are we headed what should we do and to answer those kinds of questions requires rich data and that's exactly what would be proposed here community could collect rich data about itself so it would know how it's doing and what's going to happen if i could also add one comment there with active inference type models we can have a rich uh or deep generative model even if we have sparse data collection initially and then we can ask what kinds of collection of what type of data or what type of modeling or what perturbation would make sense given our generative model or how can we differentiate between two generative models or two hypotheses at play so the richness can be specified up front we don't need to just grab what we can see in terms of data we can actually be principled in terms of what information is pursued yeah yeah yeah as we've already said part of the the concept behind active inference is decision making under uncertainty and obviously we're talking about a community that doesn't know its future and isn't sure what's going to happen next because no community understands its future very well and and especially in today's world is very uncertain about what's going to happen next will there be enough food for us next year will there be enough clean water for us next year i mean these are some big questions but there's a million little questions and so somehow a community has to make decisions under uncertainty and that's you know what the point of active inference is is how that can occur um and and then you uh touching a point that you just raised daniel is is uh you know active inference is a is a is a is a is a is a way to make steps in cognition right it's a flow of processing decision making action information flow decision making action it naturally lends itself to an ongoing process of learning and also a multi-scale one so there's a multi-scale learning here just like you said some are one-year questions one day one hour there's multiple ticks to these models that we've seen nested models at different time scales cool right right right i also want to mention too is when you know when we talk about models sometimes we might be thinking compute say computer models and that's fine you know computer miles computer models can extend the the intellect it's extension of the human capacity for processing information but you

can imagine too that individuals can individuals do process information they they learn themselves about their world and what might happen next and what they expect if certain decisions are made so you know part of part of the flow of information and the process of processing of information at the club level is is to ask people what they think will happen if we do this what do you think will happen if we enact this rule or you know put our put our uh funding priorities over on this topic uh what you know what it what is going to happen if we do that what's going to happen tomorrow what's going to happen next year what's going to happen to my grandchildren if we make this decision today i mean these are questions that humans can can take a stab at you know can answer in their to the best of their abilities and we might also imagine that certain humans are quite good at it uh you know just because certain that we all have talents in different areas so uh you know this is not just about logic and computer models this is also about engaging the human spirit and the human heart in in in directing uh path towards a better future and making decisions that lead to a better future both both in the near term and in the long term nice all right so some final thoughts to part two um uh i've i've already mentioned that it seems already that there's enough public support for this because it doesn't require a lot of public support to get off the ground again just uh you know as far as the science community goes i imagine this could move forward if just uh you know dozens or a hundred or something like that scientists were really interested in this topic i imagine it could get started and if just one community somewhere in the world is interested in perhaps doing a field trial of any new system once it's developed i mean that's that's really enough so we're talking about a tiny fraction of the world population that would have to be involved in this to get it off the and just like any new technology uh the iphone or you know like like it doesn't it once you build it and you demonstrate that it works and it brings it reduces people's uncertainty about what's going to happen next and it empowers them to make that improve their quality of life and and improve their security going forward then um you know the rest is history i think that's that's how i imagine this would work out is you just demonstrate a better way and it has such benefits that it just it just spreads so to get it off the ground just a fraction of the world would need to be involved the fraction of the population or a fraction of the of the of the global science but but once it gets rolling then i think people will actually jump out of the woodwork to be a part of this because it's like who doesn't want a better world that is demonstratively better than what we had what we have now all right so one other uh so there seems to be enough support there's there's groups uh various groups working on similar kinds of uh you know topics about say for example transformation and other topics

like that so i it's hard for me to imagine that that there wouldn't be enough support to move this forward if there were funding funding obviously is the is the big thing to get over right away but if there were funding i think this would go fine uh another another question mark i have at the bottom is you know it's a good question would transformation actually be easier than reform i mean in free form you're asking people you're asking corporations for example large corporations to change their behavior even to change how they measure success of their endeavors for example by including more social metrics in addition to just revenue metrics you're asking power you're as we'll talk soon enough you're also asking people those with those with concentrated power to distribute that power to the rest of the world and uh you know you're this is a big ask your reform requires some big actions by big players and that's great i mean reform is absolutely necessary and every effort should be made to move reform forward it's it's we won't we might not survive without reform it's especially good for the short term to get things uh you know improve conditions and improve our trajectory in the short term but in the long term transformation is necessary can i ask a question yeah um you mentioned funding being a rate limiting step so what would you spend money on oh as well as a couple slides ago i mentioned that it might cost about whatever maybe less than 10 million or around 10 million a year for the first but there has to be you know a variety of things would have to happen just to get the program going you know there would have to be conferences that would have to be some kind of staff to handle the you know the the program that would have to be you know some early studies done like the first studies would have to be funded a lot of the funding would actually just go towards studies but uh studies about what the public's opinion of transformation is what the what how transform how the idea of transformation is seen for example um simulation studies of potential to showing potential benefits like there would just there would be a slew of work that would have to occur we're talking about work by professionals of scientists and otherwise uh and uh you know that that just costs money the 10 million is actually not much so it's a small that's a small team to get things off the ground and it would cost about that there could also be funding for arts projects funding for literature funding for stories about what a transformed world would look like still you know works to to bring the idea of the potential like the potential of trent to tell the public that we actually can there's a path we can transform we could we could live under different systems and what would that be like what would that look like what would that feel like what would we you know how would that be you know that's like a it would be lots of art things and music things and literature things and media things that could support that in addition to all the science-

based uh efforts so i don't think you know there's there would be a few things you could do without funding there could be maybe a few studies that you could kind of squeak by and the spare funds or something but without without substantial funding this i don't think this would really go too far too fast all right all right well that brings us our our real topic part three nice uh so part three is focused on on design and when i say design i mean high level design concepts not not nitty gritty how would you do this how would you do that but high level design there's a few things that i talk about in this paper the the paper itself is a narrative on the topic of de novo design you know designed from scratch of new systems um obviously as we've said many times now the society is views it as a super organism and its societal systems as a cognitive architecture so in that sense the goal of design is to facilitate cognitive societal cognition through a through a better architecture i i talk i i spend time on a few topics in this paper an attention communication self-identity power and influence are all discussed and uh i also talk about uh prototypical uh societal system design the this is the leader framework that i mentioned just a minute ago i don't think we're going to get to the leader frame i'm sure we won't get to the leader framework today and there's a lot to say about that because that is as far as i understand is still the first simulation in the science literature about how an integrated system might function so there's much to say about that uh and and i don't mean to say that it uh you know the whole the whole r d program is aimed at the lead infrastructure it's not a leader framework it's not but the the those concepts could be useful developing other ideas for societal and at the very end of the paper i highlight some design considerations so we'll get to those maybe in two at least two sessions two or more sessions here cool slide eight yep okay so um i just wanted to say a few more words about you know reform versus transformation so uh there as i mentioned there are loud voices at the u.n and even in governments national government state governments local governments and certainly in the general public and certainly in the science community calling for bold really radical change to avoid this cliff that were hopefully not going to fall over here related to climate change related to biodiversity loss and a host of other uh social and environmental problems and um you know just to just to kind of emphasize how how dangerous our situation is it is not clear if humans will go extinct with all the other extinctions that are happening and i don't mean extinct in a thousand years i mean like will there be will there at least will there be a civilization any any sensible kind of any any civilization that we're used to will that exist in who knows 10 years 20 years 50 years 100 years you know like our our future is highly uncertain because we have not addressed deep problems that have been with us for for many decades we've allowed them to grow worse and

now we are all at risk at risk of just lower quality of life or risk of mass die off of humans and at risk ultimately of human extinction so there's lots of voices calling for bold change and that bold change can be seen in a couple of ways more the typical way to see it is reform of uh isolated societal substance systems like uh there's a lot of focus if you just read the news there's a lot of focus on green energy for example using solar powered energy and things like that uh better trans you know electric cars rather than gas gasoline engines that's all great again all of this stuff is wonderful but i would i would offer that that there's something really missing from this kind of assessment and that is the larger picture the the the the meta level uh they're sometimes called in the mlp of the multi-level perspective gills it's called the the landscape level or the regime level and i would say that we have to address that level if we're going to really solve our problems if we're really going to develop if our systems are really going to act well as cognitive systems then we have to address the systems level the parameters and characteristics so right now the focus of all all the work that's being done in sustainability and this bold the idea of bold transformation almost all of it is focused on this kind of meso in between level uh you know the say the energy the energy infrastructure or the transportation infrastructure and this in this whole series instead is is really focused at the at the landscape level the the large the whole transformation the whole systems level okay i just wanted to make that point clear and okay so now now i you know we're we're already in part three of the series and i've i've i've listed six systems that i that are six you know meta systems or landscape level systems that i've that i'm interested in economic political legal health science and education but in the first few pages of this uh paper i i finally say what that actually means so uh so uh the i'm basing my definition of a societal system on what's called a function system and the more there's citations there by roth and schultz and others about what a function system is comes out of social systems theory but they list uh three characteristics of a function system it has to have be observable and not a subsystem of other societal systems so the economic system is a you know entity in and of itself not a subsystem other other systems um involves a communicative system whose operations refer to a society and manifests the functional perspective at work at work so uh roth and schultz identified ten function systems and uh i would say that uh six of these i'm calling the the overarching systems the economic political legal and so on in in parentheses there i have their functions so the function the one word function of the economic system is distribution the one word function of the legal system is standardization the health system is restoration and so on the education systems formation there's a four other uh of function systems that roth and schultz identify uh art religion sport

and mass media and i do talk about these but i talk about them more indirectly so that's what i mean by a function and and i also use the word or that's what i mean by a societal system and i also read the word institutions in throughout the series and it's worth spending just a little bit of spending a little time describing what i mean here um i use that in the term in two senses one is just concrete organizations for example a attacks or attacks uh institution inside of a governance system for example but the second meaning is is more broad it's uh more more along the lines of a social institution and this is kind of interesting because grabner and gobarney define social institutions they give a little like a little blip there codophile systems of social structures and particular norms and rules that incline individuals to act in specific ways and uh interesting the idea this is you know they've developed these ideas apart from active inference but the but the language is not so dissimilar because they talk about institutions as constraining and enabling people to act coordinate in complex and uncertain environments they stabilize the expectation of of individuals about the behavior of others and uh thus the institutions allow individuals to offload some of the cognitive burden that that they would otherwise be required to compute if if there's some kind of social norm and i expect you to behave in a certain way under a certain certain stance i don't have to spend a lot of thought figuring out what you're going to do next as i go about my life you know i can i can just assume that you're going to act according to the norm and then that's a whole lot of work that i don't have to do to to forecast what i think you're going to do so uh institutions allow individuals to offload some of the cognitive burden and uh and and anticipate the state of and the beliefs of others so obviously that's that's related to active inference right active inference is uh uh at the you know when we talk about uh uh um of forming you know structuring a society or structuring a society to reduce one way to do that is to develop a set of norms that this that everyone agrees to where everyone follows or mostly follows so that the uncertainty about what's going to happen next is reduced so so active infra developing institutions if they're quality institutions actually helps them to reduce free energy and expected uncertainty i think there's a few dimensions to this and it will be fun to unpack with what does it look like to have the active inference institution what does that mean as far as the operating procedures of the internals of the business versus how the stakeholders interpret it how it's communicated so how it actually plays out will be something we're just exploring yeah yeah and when i say institutions i'm you know i'm talking uh widely so financial institutions uh you know social institutions cultural institutions and so on um and i'm i'm suggesting that they can the institutions can be viewed through the lens of active inference and may be improved by

understanding you know how human cognition works and how group cognition works we might be able to improve our um just some reminders from from our previous talks about the goals and purpose of the r d program and of and of the purpose of an organism itself and thus the purpose of a of a societal system that serve that serves a group of organisms the society so the the goal of transformation from this view is to develop and implement systems that can best facilitate societal cognition so that's the goal we have metrics to assess whether a new design is better or worse than some other design and importantly if if we weren't thinking about cognition if we were just thinking about conditions we might say the goal of transformation you know improve education levels or improve diets or improve quality of life in some way or improve economic security or any any number of things like that all of which are fantastic are good but once we bring once we bring this new idea of of of societal systems as cognitive architecture and into play into into play then we're talking about um the the the conditions such as clean air or clean water being the outcome of a decision-making process so the focus starts to shift to the primary uh you know the the primal goal of quality cognition versus the secondary goal of what quality cognition brings quality cognition at the societal level brings clean air and clean water and low uncertainty and you know higher you know good education and freedom and you know any number of other characteristics conditions you might want to list so it is important that you know this is this the by looking at uh societal uh systems as a cognitive architecture suddenly we have a much richer understanding of of not only how a society works and how decision making in a society works but also we have new metrics by which we can measure the success of a societal system right and so this would include cogniti cognitive metrics data metrics information metrics processing metrics you know all kinds of so that's a little bit on on the goals of transformation and the goals of the and a little bit on the purpose uh the purpose of the program and the purpose of a of a new design right i just want to say the purpose of a design is both aspirational and functional we've already talked about design as the last step in the process of world view leads to purpose leads to fitness fitness metrics and then finally to some kind of system design so in that process in that process obviously uh purpose is uh is functional like you know everything else practically everything else depends on purpose you know or many things depend on purpose but beyond that having an explicit purpose for a club say it you know like for a particular a clear statement of useful purpose can actually help in and of itself to a shared purpose can help to unite individuals and focus attention and uncommon causes and encourage cooperation if we all know why we're doing this together if we all know why what we're doing and why we're doing

it and that and the and that purpose is to improve our collective quality of life and improve the environment well then it's going to be a lot easier to get people to come together and talk and and get and you know unite behind a shared purpose uh so in this sense a purpose itself could aid societal cognition right societal cognition depends on cooperation depends on trust so just having a just having a purpose can aid in societal cognition and and you know like as a as a counterpoint to that what is that what is the purpose of our current society you know we have these native systems of governance and economic systems you know so what is the what is their purpose right what is the purpose of an economic system you know i mean we could we could debate that right but but there it's i would say that it's uh that if you just take things at face value one purpose would seem to be to promote revenue and sales and consumerism and if that's the purpose if that's part of the purpose of an economic system then why would you know it's hard to get behind that you know it's hard for a uh for for a whole society to say oh that's a that's wonderful let's get behind consumerism because it's so destructive okay um so a little bit on goals and purpose and on to the next slide uh this is just a quick review of active interest maybe i don't even need to say this because we've gone over it a few times but essentially uh you know the short view is that we can we can view action action cognition as an uncertainty reducing process so that's that's what what this is all about in a nutshell and uh the the uh through via via the active inference mechanism we take in observations we update our models we learn we take actions that can reduce our uncertainty either either either explorative actions that bring in new information epistemic epistemic actions or goal-directed actions that actually immediately or in the short term or act to improve our situation so it's all about reducing uncertainty and especially it's about reducing the uncertainty of essential variables and this is of course uh you know this is a rich topic in and of itself what are the essential variables of a society i mean they you would imagine that in some way they would include clean water and food and you know sufficient food clean air and shelter and all those kinds of things but as we talked about earlier in the last discussion there's we talked maybe eight or so core human needs that include affection and freedom and a bunch of other core human needs so we're complex organisms and we have complex needs and these these essential variables the variables that really matter are themselves a complex set of variables and if we and if we focus our attention on on addressing this question of uncertainty about essential variables that almost means by definition we're doing what i what i like to call solving problems that matter problems that matter are the problems that really relate to the essential variables about how are we really doing how are we going to do tomorrow how will

we do in 10 years yeah so so i just we'll just read the last the last sentence of the last block of text i have there this is by bad and they just they phrase they're speaking about active inference and uh the act of influence as it plays out at the social institution level and they say that the elegant idea that we can operate together to minimize collective stands to cast a new light on classical social phenomena and that's about it in a nutshell that's it and to draw one point on that in contrast to for example reinforcement learning based approaches or reward maximization based approaches it's almost like you have the gaussian distribution it has a mean and a variance and the reward learning is like let's shift the mean up as much as possible i mean isn't the average income right now in us dollars higher than ever well maybe but then if you think about what the dollar means and how the dollar is distributed then the average doesn't really tell you too much about the distribution so active inference is about reducing your uncertainty on a specified generative model so somebody might think i think the distribution we might want to see this type of distribution this shape or this kind and that's a discussion to have and then we can reduce the variance around that and let the mean fall where it may so it's very different to have a reward highest as possible because having high expectations but in unstated and often unformalized variants can lead to some ridiculous outcomes like the saint petersburg paradox and other uh investing fallacies let's say yeah good point um uh so so your point was that uh uh reinforcement learning is aimed at not it's not really a necessarily a probabilistic uh approach and and if you and you will have to really ought to include the distribution of whatever targets you're looking at in order to get a better idea of you know of what's a good decision and what's not a good decision absolutely another difference between rl and active inference is that the programmer that whoever is designing the experiment or the whoever is coding the robot or whatever has to decide in advance what the what the rewards are for certain actions you know you fall through trapdoor negative 10 points you you you you get the you win the chess game you know positive 10 points or whatever the game is and uh active inference allows the the agent to learn goals as it goes uh learn the world as it goes and make to change its change its priorities as it goes which is which is exactly what humans have to do we have to be learning as we go and changing our perspective and changing our what our goals are and how what we're valuing we have to change that on the fly in order to maneuver in a highly uncertain world if i could add one more point on that it and this was expanded upon by uh noor sajid and philip ball in our one of our previous discussions that relationship and the learning of the cues in the world can be basically semantic and arbitrary so sugar isn't just always simply a reward for the human in

the like right now i don't feel like having sugar so it's not part of my preference vector so with active inference type models we can separate out okay what's on the table in terms of our current affordances how about future affordances what are our current preferences and how sure are we about them and all these parts of the actual action plan that will help us align on shared action rather than thinking that if we simply line on shared thought then all of a sudden we'll know what to do because that's not even true right right good good okay so um so um we're we've been talking about societal systems as a cognitive architecture all through this but what is maybe it's time we actually gave a little thought to what cognition actually is right um so uh lyon lists 13 cognitive tasks that he that that he says qualifies qualifies as universal tasks or capacities and uh it is ra you know it this is this is part of the world view that we we talked about in the first uh in the first discussion in that the all of life is intelligent all of life is engaged in cognition all living things are engaged in cognition and these are some of the you know some of the tasks and capacities that what we mean by that so this these hold true for a human certainly but they also hold true for bacteria or slime mold or plant or any any other living organism i i don't know if i need to go through all 13 one by one but i'll just kind of jump around a little bit so just number one there has to be some sort of repulsion or neutrality or some kind of judgment of what's good and what's bad what's useful and what's not useful certainly to cognate you have to sense you have to bring in information there has to be some kind of memory storage or some kind of recall there has to be some kind of learning some kind of problem-solving behavior there has to be communication with other agents of the same kind or different kinds there has to be anticipation certainly we if we're going to make a decision about our future we have to anticipate what our future might be if we do or don't do some kind of act um there's attention there's a million things to spend our attention on as a human and what we spend our attention on will govern in large part the quality of the decisions that we make and the last one i'll talk about is self-identity so you know if i say i'm going to the store in that sentence i'm talking about this particular body right and my self-identity is focused on this particular body but i can also talk about myself as a citizen of a community as a citizen of a nation of a of a as a part of an ecosystem as a part of the biosphere so we have the capacity to self-identify at numerous levels i call that extended extended self-identity you know not just ourselves maybe our family maybe our community maybe our environment maybe our local environment maybe our global and maybe of human civilization we're a part of human civilization and as we'll see shortly how we how we self-identify governs to some degree the kinds of decisions and the kind of kinds of behaviors that we

display so uh so these are these are we can think of these as the you know occurring in every organism and the next slide brings the two ideas together the idea of function systems and the idea of these universal cognitive tasks how do they relate to each other and i'm offering that that in a sense these 13 cognitive tasks are really you could think of them as building blocks for the higher level functions so for uh an economic system say in the in the function view an economic system might involve all 13 tasks in one capacitor or another a education system might involve all 13 tasks but might particularly involve the tasks about learning for example so we can think of a lines list as containing low level cognitive building blocks and function system as being the higher level functions built on top of these or from these and then we can think of active inference as the multi-scale process that connects the two and these function systems then provide a cognitive architect architecture that provides a society with a with a cognitive flexibility that is a society needs to make it has these uh you know six or ten different uh function systems that it can make decisions through that it can learn through and there would be certain kinds of problems that are maybe better suited for say an economic system or a governance system or something and other kinds of problems that might be better suited for a legal system or education system but maybe some problems really have to engage multiple function so the fact that a society has these various uh systems of cognition it gives it a certain cognitive flexibility flexibility and provides a certain kind of redundancy and and then obviously raises the question of what kind of balance are these you know eight or ten function systems you know how does the dynamics of that balance play out uh next slide okay in this slide i'm just listing the next four things we're going to talk about and that'll be the if we get to it that will be the end of this our discussions today um those are those are neuronal versus human societies what are the similarities what can we learn from neuronal societies that might help us when we're thinking about design all of these all of these four points are related in some way to the design of new societal systems influence how do we influence each other and particularly what is the role of coercion and influence self-identity and selfishness we we just were talking about that when i was talking about expanded self-identity and we'll pick up that theme when in a short discussion about cancer what is cancer and then lastly is power distributions we have we have eight or ten or six or ten functional systems uh if one is you know if one of those is unbalanced with the other two has too much emphasis in the society than others how does that affect a society and then also within a system if for example in an economic system if uh financial power is is uh you know concentrated how does that affect a society so we'll talk about that all that and and then with the idea that all of these pertain to the

design of new systems in some way we can learn something from these topics about how we might want to design new so slide 16 um so it's kind of interesting you know our what the associations and community that happens inside of our brain to for human to solve problems is not so different in some ways about how we engage with each other in in potentially solving problems as a society so uh i'll just kind of read a little bit from that first paragraph but similar to neurons individuals connect with each other in complex ways just as neurons do we exchange a variety of signals over a variety of distances just like neurons do at a variety of strengths both neurons and persons can potentially connect to a very large number of others each can share each individual each neuron each person can share multiple connections with others and these connections can form meta level structures like institutions or tissues or whatever the case may be neurons and persons self-organize into modules of tissues families clubs businesses etc and neuron neuron and person-person connections are dynamic and context and need dependent the the neuronal architecture is quite just as the human human social interactions are flexible associations come and go even if there remains some kind of large scale you know consistent structure and and each individual is cognitive just as each neuron is cognitive that's actually kind of a new new new result in the science literature the capacity of a neuron to process in and of itself so uh you know obviously we want our neurons to work together we want them to associate with each other in a way that helps us to solve you know our particular problems we want their signals to be strong and good so that information is moves throughout the you know the neuronal process as it should uh we want to do we want our neural neural architecture to re-reorganize itself as we need to as we need to learn new tasks for example or new topics and all of these are this is also what we want for societies we if we're going to be part of this cognitive architecture if we're if we're engaged in societal cognition we need to associate freely with each join join the collaborations and groups as we see fit that's part of societal cognition um our you know societal cognition in particular the designs of societal systems might impact any in this bulleted list that i have but one the first is the strength and the quality of associations between individuals the richness and the extent of our communication uh for example a a design a societal system design might as part of it a means of communication so that say a large group of people can get together and collaborate and coming up with a decision on some topic right so the the societal system itself could communication and coordination aspects to it temporal and spatial locations and movements of individuals affect our societal cognition and and designs can affect that uh quality of information that is gathered uh and in particular i'd like to point out the motivation to engage

in say economic behavior hopefully will be different than it is in native systems where the motivation is largely to you know personal uh income and wealth um uh we would want go ahead oh just one one uh i was gonna ask what the differences were in contrast but finish whatever you're going to go ahead and say oh well just on that topic you know a new design would want to employ a set of motivations that is you know very useful for societal cognition and societal health and certainly encouraging people to act in a kind of a selfish self-centered strict strict selfish way for their own gain would make little sense you know that's that's like a design flaw in our current system you know that we're not encouraged to cooperate and encourage to solve common problems together and the design of a system can deeply affect what our motivations for actions as a matter of fact a design ought to make cooperation so easy that it becomes the de facto default behavior it should that should be the kind of you would think that would be the how it how a good design would work is it would just reduce the barriers to cooperation and communication and coordination because th those are all key to um societal decision making and learning yeah uh other just a few others here uh the design might affect uh for example how well as a society's beliefs aligns with its reality you know for example if in a new design there's a strong role for science and there's funding in you know funding in clubs for scientific exploration well then maybe it's more likely that that new systems will be more aligned with the beliefs of people working in new systems or engaging in new systems might be more aligned with realities because you know the best science has it best as at its best is aiming towards a better understanding of of of of our world um design might affect how independent co-op cooperative individuals are as i mentioned a minute ago um also might affect how uh well how deeply the level of critical critical thinking is right how how well does a society does a community encourage critical thinking through education and through norms and through other processes but obviously critical thinking is important in problem solving you can't solve problems if you're not doing critical thinking you can't solve problems well and and also the capacity to plan the extensive and the extent of the anticipatory horizon you know what a design might affect uh how how much attention is paid to various predictions made by organizations and groups um maybe it would be important to pay attention to the predictions that people made so we can understand who is good at who is who has a skill uh you know who's good how can we improve our capacity to anticipate what's going to happen so in numerous ways in the same sense that design design of a brain system might affect cognition within the neural communities some of those same uh ideas can play out in the design of societal systems because you know associations and etc etc would be

important at the societal level i'll just point out as an evolutionary biologist that selection has shaped on stumbles certainly and so several patients of today have benefited from developmental basically patterning that has been selected on for a long long time many many many generations so real evolutionary pressures the kind that create resilient neural systems at the level of societies would require essentially population-like dynamics competing and the birth and the death and the diversification of human societies so systems mapping is really helpful and the toolkit transfers there for sure but there's a key evolutionary bit that neuronal ensembles have that ant colonies have that simply saying distributed systems are like those two things uh is is not there but in the long range that's where it would be yeah yeah um i i think i've said this before but um but you know if we're talking about how can society do better at cognition be you know what kind of systems might facilitate cognition well then all we have to do is look at evolution we look at how the what what is the world doing around us how does a brain work these are you know the the the information and the ideas and the and the and the patterns that we see throughout nation of throughout nature can be look we can learn from them as to how good cognitive systems work because nature is built that nature is a good cognitive system right there's so much to learn from nature cool um all right so this is the second of the four topics uh how do we influence each other so we have a group obviously the people will there's problems to solve there's activities to to undertake there's uh challenges that arise and group members would try to influence with one another in some ways through communication or you know some kind of communication and turner the fellow i'm focusing on here uh his ideas come from uh he did work he worked in uh social identity theory self-catering categorization theory and some other related fields and he uh identifies two forms of influence now this is not this this is not coming from you know the active inference world but i'm going to make the point at the end that the two are closely aligned with each other um one time type of influence is persuasion and that's where you simply argue that your you know your understanding of realities in some way better than other understandings or your your path of action is in some way been more beneficial than other paths of action but already then we're you know we're talking about aspects of of cognition in the human brain or active inference because we're talking about some kind of uh testing of a model you know of of looking at potential paths forward in action and then evaluating them as for as regards to which is the most beneficial path to take so already we're we're touching on the ground of active inference another form of influence is control and that comes in two flavors legitimate authority and coercion and legitimate authority we've already talked about this because we've talked about

institutions and offloading of cognition from the individual level to the group level through social institutions of one kind or another norms or actual concrete institutions and things um and by and large legitimate authority people people are are are willing to uh prove to you know accept legitimate authority that because they understand that it's in their best interest you know one one example is we most of the work most of us in the world are okay if uh if a traffic cop stops someone who is uh you know speeding at a very high speed because that's dangerous to us all and we are glad the traffic cop is going to take some action and exercise authority over this fellow who's endangering the rest of us so by and large we we we are good with legitimate authority and we don't feel that it limits our our freedom um and then there's coercion which is more of a confrontational uh form of influence where you maybe force someone to to take us behave in a certain way or to um you know act in a certain way and there's various forms of coercion that can occur uh including various forms of manipulation i'll just list a few but that could be disinformation deceit emissions obstruction the exercise of false authority drowning out or silencing of competing voices or various things so you know this is rampant in our societies today these these occur you turn on the tv and you'll see evidence of all of these and uh turner makes the point that coercion all of these forms of coercion is easily abused and very dangerous to a society because it re that tends to reduce trust in societal institutions it tends to reduce the long-term health of the group can be destructive in the sense that coercion breeds resentment to you know to being forced to behave in a certain way and it can reduce trust and reduce cooperation and it increases an antagonistic climate and uh that remains then that the society who's during the coercion of the group that's during the coercion has to use even more energy and more resources to buckle down and make and make the question even more severe so that they can get the individuals to behave in the way they want so uh you know there's a role for coercion especially say like a you know police uh uh arresting a someone dangerous who has a gun and is about to shoot someone you would hope that they would tackle that person to the ground or whatever and prevent them from doing that from hurting others so there's a role for coercion but it can be easily abused as i said and must be used very transparently and judiciously or else in the long run the whole society is going to suffer and potentially go down the tubes um so that's the way he sees it and then we can we can see how this plays out in the active active inference framework already we've talked about persuasion and the the similarities between persuasion and learning and then legitimate legitimate authority as being similar to offloading of cognition to insta social and obviously when we're talking about influence we're talking about group associations and

we've already now mentioned that in with respect to neuronal associations and how the how neurons and cis and people in a society have to have the freedom to associate at will to to provide flexible cognition for the um right so association is a part of just the fact that we're associating and and and uh you know trying to influence one another is a part of cognition so so what are the impacts of this on design like how to how would we how would how might we change a design or or what kind of design elements might we incorporate into new societal systems if this was our concern if if we wanted to encourage persuasion and legitimate authority how you know what this is the question i'm asking you know the world in this paper is how how does how could this impact design and you can think of a few things off the top of your head for example transparency would help this uh coercion is very difficult to achieve if there's high transparency if there's quality information flowing then coercion is more difficult to achieve if there's quality education happening maybe even especially uh education about what coercion is what what is manipulation what are the what are the many faces of manipulation how do different groups in society use how do they how do they uh how do they manipulate in order in order to you know get their will how is uh how what is the role of media in coercion you know these are all these are all good questions to ask if we're talking about what kinds of new designs would we want in order to avoid some of these problems the next of these four topics is self-identity uh and uh use the example of cancer here so uh um levin has published a couple of papers that i think are really interesting on this topic and uh it's it's cl it's true that as cancer as a cancer develops in a body the communication tends to up to be changed there there can be less quality communication between a cancer you know between a tumor and the surrounding tissues and at the same time there can be more communication between cancer cells or within a tumor so so communication kind of goes haywire in a cancer or can go haywire and uh the the idea that levin has is that what happens is we often we often we often portray a cancer as being a selfish you know a selfish growth that's occurring in the body and he says you know it's not really quite like that it's uh it's a little bit more like both normal tissue and cancer tissue is self-interested and acting in a self-preserving manner the difference is what the extent you know the functional extent of self is for the two populations so the functional extent itself for for normal tissue normal cells in a normal tissue is the whole body like the cells are connected and are communicating with other cells in the body both locally and distantly so the the body is a cognitive apparatus you could say an extension of an individual cell outward to a larger cognitive apparatus a larger environment a larger milieu and in a cancer cell that those communication that communication is breaking

down and this the functional extent of cell for a cancer cell might be you know might grow smaller and smaller and smaller so the cancer cell just like the normal cell is acting in a self-interested and self-preserving manner only its functional extensive self is small so uh it's taking care of us it's including in its self a small group of cells small tissue as uh relative to the body so it's it's its ability to to think in a sense cognate shrinks because it's no longer attached to the rest of the cognitive apparatus of the body it's has to do any cognitive processing that it can do within a much smaller set of cells so it's capacity to act say over multiple time scales to behave in such a way that it will exist in two years or five years is reduced because it's it's he has a nice term for it he calls it the cognitive light cone of self is reduced in the cancer and this is really interesting you know when we talk about us how this might play out it's at a societal level right there's uh you know there's there's maybe some injury or stress in a certain that population is retracting itself from maybe an from the rest of the society the rest of the environment because it's maybe feels threatened or it feels unheard or it feels unappreciated or if just is under stress and uh it's you know the the local group is forming that is more interested in the local cognitive uh light cone of self a smaller like cone of self uh and that can be dangerous you know if if that happens too much in a society that can be dangerous it can also be useful because people are getting together in small groups to think about how to solve a problem that can also be useful but but if that is if there's too much of that or if it happens under the wrong circumstances then you can get the situation where the society is really kind of starting to disintegrate because because there are elements in the society that are acting preserving the small self and acting to you know protect the small self as opposed to the whole and you know you can just imagine what that leads to and so the so the task of design then is to create a a cognitive framework that where in which communication is rich in which everyone's voice is heard in which problems are actually addressed and not just glossed over or ignored where where the needs of uh you know the whole are taken into account uh where everyone's story matters where everyone has an opportunity to impact the larger workings of the society the decisions that the society makes and uh you know building such a design building such a and a cognitive system then would reduce the risk of you could say cancerous growths occurring uh right so uh you know the cancer is informative in this sense you know it's an amazing i what my background is in cancer biology and it's a never-ending source of information and you know to look at how complex the complexity of the situation surrounding cancer so that's it so you know we can design systems so that cognitive light cone itself is expanded rather than contracted and that's

going to go a long ways towards towards you know kind of encouraging cooperation encouraging coordination encouraging group problem solving on group problems you know nice thank you the last topic is uh power distributions so we've been talking about uh functional systems each functional system has a function i've listed some of those and each functional system also operates via a particular medium so for example you know the a good example is the economic system operating under with the through the medium of money uh so some authors that i cite suggest that each functional system if left unchecked tends to have a growth imperative and we certainly see that in the economic system it's like everything is now suddenly uh my couch and my living room is now worth money because i can sell it to someone who wants to visit my hometown for a weekend right so everything is commoditized now or many many things are commoditized that used to be more freely shared without just in the course of life with uh with people who had needs so i sell my i sell my time i sell my resources i sell my ideas and uh and and and financial the financial system seems to have overtaken many aspects of our life so in that sense in our current society you might say that the the financial the economic system is over emphasized relative to the other functional system say the education system and other systems so uh obviously then when we're talking about design one of the goals of design is to balance the the the influence of each function system and that is a topic that we already mentioned very briefly about in integrating systems together so integrated economic system legal system governance system you know all serving the same purpose and all acting as a as a flexible uh cognitive architecture for the greater good of the of the whole if i can give a comment on that uh you mentioned checks and balances and that in the american context referred to several separations within the governance mechanisms and the development that happened subsequent here we're talking about checks and balances in a little bit more of a general cybernetic goal-oriented system way and we're talking about checks and balances between these different sectors it almost sounds like so it's not just that there's going to be market mechanisms with all of the manufacturers of a product competing on that one dimension and then on a separate sector there's competition of the academics on this one question in this one way how can we link those competitions in a way that actually is aligned with the overall goals of society right exactly how can we align all of the systems to which is the the you know the the reduced uncertainty and expected uncertainty of the of the society in relation to the essential variables the things that matter right so that's one uh aspect of design is how can we maintain a balance among but another aspect is how can we maintain a balance within systems so the economic system again provides a really good example here

and i i say i have a sentence there that today the wealthiest one percent of individuals globally own about 40 percent of the world's wealth now that is like insane i mean that is crazy especially if we think of money as a voting tool i mean money is a voting tool it's it's an info it's not a recognized formal voting tool but it's a undemocratic one if i have if i'm a billionaire i have lots of votes in this world i can get people to do any number of things that i wish them to do i can direct non-profits to focus on particular issues i can i can make a new company that does a particular you know provides a particular service or or you know i can i can i have all kind i have a billion times more influence than just a normal person right so money is a voting tool and it's just uh it's just used as an undemocratic one but it could be used as a democratic voting tool it could the money system the economic system could be highly transparent it could it could achieve a um a reasonable degree or even a strong degree of income and wealth equality so that the money as a voting tool becomes a fair voting tool right like in a democracy we would not we would not stand for i get one vote and some and the guy down the street gets a billion votes i mean that would just we would recognize that as being deeply unfair but it's the same thing in an economic system if if one person has one dollar another and the next guy has a billion dollars that person has enormous control over how uh society makes decisions so it's not just economic power i mean i'm focusing on the economic system but there's you know there's six or ten systems that we could be focusing on here and each of them has some kind of medium and we could talk about the distribution of power in any of them for example we could talk about the distribution of power in a in a governance system currently in the u.s there are what 400 i don't know how many how many legislators and senators do we have in total i don't know what that amounts to but compared to the population this is a time we have a we have empowered we have put enormous power in the hands of a tiny sliver of people and you know does is that a good in today's world could we do better i imagine that a different kind of governance system would destroy would do a far better job of distributing power so that everyone can have feel like their voice is heard and that they can be part of the decision-making process you know as it stands right like right now in america i get i get to vote every two years uh about like a yes i get like a thumbs up or a thumbs down on certain candidates and that's that's kind of it formally that's it i mean i can do some informal things i can write to my congressman or i can you know i can complain in the newspaper or write a letter to the editor or something but as far as formal things that i can do to have my very nuanced understanding of the world heard all i all i really get is a is like a yes or no every few years you know and then a group of candidates and that's hardly that's hardly

taken advantage of my understanding of the world right i i have much more uh insight to offer than just a simple yes or no every few years so i think it would be possible then to to to design systems that balance this medium of communication in each of the function systems achieve some much greater degree of incoming wealth equality achieves a much greater degree of power distribution within governance systems uh power distribution within legal uh you know there's a great room for improvement on all of these uh yeah so i think is there one more oh there's one more slide yep and uh that is really focused on economic power so the first one i already talked about using money as a voting tool we already do that and we would be better if we just did that above board and in a democratic and transparent fashion and new systems could do that and when i say economic systems again i am including in that monetary systems financial systems and the bigger picture um a large set of studies have shown that incoming wealth inequality is bad for society it affects uh health cooperation trust anxiety happiness social cohesion and then mortality and uh social stability in a number and numerous other factors so it's quite clear that high amounts of uh any income and wealth inequality are detrimental to society and and again today we are extremely uh unequal extremely unequal and then uh the last just to end this whole uh talk today i would like to point out that income equality is really not a there's a it's kind of it's not unpopular the concept is not unpopular there was a large study done a few years ago where they showed they didn't they didn't they showed participants uh graphs of the division of economic power income and things like that uh two you know they had three different choices an image that was really uh very equal like 50 50 equal one that was uh fairly equal but not quite you know not quite perfectly equal and then a really extreme one well it turned out that people didn't you know they asked you ask people like which of these do you prefer and it turned out that they didn't like the extreme one at all and that happened to be the one that was the picture of the u.s inequality in the u.s and uh and the one that was more mild i think was the inequality maybe in in sweden or something like that so they they preferred the the distributions of economic power that were less less extreme and when asked which of these do you think is the most fair for the society the the the uh more than half of those queried and this was many thousands of people who were queried chose actual income equality like you know like completely in equal incomes as the most fair way to run a society and i don't i don't suppose that they that people really thought that was possible like you know that's i know it's not possible but that would be the more fair way that's what i imagine people were thinking and uh when we next come back to talk we'll talk about the leader framework and some potential designs that actually would achieve income equality

income and wealth equality over time so that's that's yet to come wow awesome that really flew by that i uh after some little tech bumps it was awesome and that is a little bit like an analogy for maybe this program there's some technical details up front few live streams to start a few few teams few memes but um maybe once there's momentum and once there's proof in the world and a little bit more of a contact between the theory and the practice we would see some new kinds of momentum so maybe give a last comment what is going to happen in the fourth session and then our two group discussions and um yeah yeah we're gonna work we have one more uh i'm glad you said that because i did want to say to folks that we have one more discussion in this series coming up next week and in that discussion we'll talk about the leader framework as a prototype and some of the things it might achieve how it might work and that is a big part of our discussion next time and then there's two regularly scheduled regular scheduled discussions on part three of the paper that will happen in may i think the first one is in two weeks it's the first two tuesdays of may at 7 00 a.m pacific and that will be like the group active inference live stream so we'll have a bunch of other colleagues and that will be more of a group discussion but the fourth conversation between just us will be april 27th at 3 30 pm pacific so i'm really looking forward to that and thanks for um for coming on for so many times and giving us so many good insights and we appreciate your time and all the participants who stuck with us also for the tech thanks thanks for having me after